

UChicago REU Notes Week 5

Zih-Yu Hsieh

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The goal for this note is to get a hand of what 2-categories are, both intuitively and some tangible generalization.

1 Motivating Examples: Natural Transformations and Homotopies

First, let's consider the most explicit example, the category of small categories, $\mathbf{Cat}$:

Example 1.1. Given two small categories $\mathcal{C}, \mathcal{D} \in \mathbf{Cat}$, the homset is the *functor category* $[\mathcal{C}, \mathcal{D}]$. Which, the morphism in $[\mathcal{C}, \mathcal{D}]$ are the natural transformations between functors.

To keep in mind that we're still doing topology (in most of the cases), let's provide the category of topological spaces $\mathbf{Top}$ as an intuition:

Example 1.2. For each $X, Y \in \mathbf{Top}$, the homset $\mathrm{Hom}_{\mathbf{Top}}(X, Y)$ is in fact a groupoid: Define morphisms between maps $f, g : X \rightarrow Y$ to be homotopy classes of homotopies (denoted as $[H] : f \Rightarrow g$), with compositions being concatenations of homotopies: If $[H_1] : f \Rightarrow g$ and $[H_2] : g \Rightarrow h$, then $[H_2] \circ_2 [H_1] := [H_2 \cot H_1]$ as follow:

$$H_2 \cdot H_1 : X \times I \rightarrow Y, \quad (H_2 \cdot H_1)(x, t) := \begin{cases} H_1(x, 2t) & 0 \leq t \leq \frac{1}{2} \\ H_2(x, 2t - 1) & \frac{1}{2} < t \leq 1 \end{cases}$$

Then, $\mathrm{Hom}_{\mathbf{Top}}(X, Y)$ forms a groupoid (a category with all morphisms being isomorphisms).

If associating each small category as topological space, functors as continuous maps, and natural transformations as homotopies, the first example is analogous to the second one.

An intuitive picture one can have in mind is: In 2-categories, objects are 0-cells, 1-morphisms are 1-cells, and 2-morphisms are 2-cells (serving as transformations between the 1-cells).

2 Enriched Categories

This section follows Emily Riehl's *Categorical Homotopy Theory*.

To properly define notions needed afterward, we need a concrete way of viewing categories enriched over small categories. This more generally follows categories enriched over symmetric monoidal categories, which are categories whose hom functor (or some similarly-behaved functor) are endowed with extra structures relevant to a monoidal category.

Notation 2.1. We'll use $(\mathcal{V}, \times, *)$ to denote a symmetric monoidal category here. If monoidally closed, we'll use $\mathcal{V}(-, -)$ to denote the internal hom functor.

Definition 2.2. A category $\mathcal{D}$ is *enriched over* $\mathcal{V}$, or called a $\mathcal{V}$ -category, if it consists of the following data:

- For each pair $x, y \in \mathcal{D}$, there is a *Hom-object* $\mathcal{D}(x, y) \in \mathcal{V}$.
- For each $x \in \mathcal{D}$, a *unit morphism* $\text{id}_x : * \rightarrow \mathcal{D}(x, x)$ in $\mathcal{V}$.
- For each triple $x, y, z \in \mathcal{D}$, a *composition morphism* $\circ : \mathcal{D}(y, z) \times \mathcal{D}(x, y) \rightarrow \mathcal{D}(x, z)$ in $\mathcal{V}$.

such that the following diagram commutes when given all $x, y, z, w \in \mathcal{D}$:

- *Associativity of composition:*

$$\begin{array}{ccc} \mathcal{D}(z, w) \times \mathcal{D}(y, z) \times \mathcal{D}(x, y) & \xrightarrow{1 \times \circ} & \mathcal{D}(z, w) \times \mathcal{D}(x, y) \\ \circ \times 1 \downarrow & & \downarrow \circ \\ \mathcal{D}(y, w) \times \mathcal{D}(x, y) & \xrightarrow{\circ} & \mathcal{D}(x, w) \end{array}$$

- *Existence of Two-sided Units:*

$$\begin{array}{ccc} \mathcal{D}(x, y) \times * & \xrightarrow{1 \times \text{id}_x} & \mathcal{D}(x, y) \times \mathcal{D}(x, x) \\ \searrow \sim & & \downarrow \circ \\ & & \mathcal{D}(x, y) \end{array} \quad , \quad \begin{array}{ccc} \mathcal{D}(y, y) \times \mathcal{D}(x, y) & \xleftarrow{\text{id}_y \times 1} & * \times \mathcal{D}(x, y) \\ \downarrow \circ & & \swarrow \sim \\ \mathcal{D}(x, y) & & \end{array}$$

These conditions are mimicing the idea of homsets in a given category. For instance, with this definition by choosing $\mathcal{D}(-, -) := \text{Hom}_{\mathcal{D}}(-, -)$, we immediately realize all categories are enriched over **Set**. Since more general categories don't have concrete descriptions of objects using sets, we instead need a generalization so that the analogy of "hom functors" can land in more general categories.

Example 2.3. Given a one-object category μ enriched over $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$, the unique object $*$ has a unit homomorphism $\text{id}_* : \mathbb{Z} \rightarrow m(*, *)$, a multiplication homomorphism $\cdot : m(*, *) \otimes_{\mathbb{Z}} m(*, *) \rightarrow m(*, *)$.

Denote $1_m := \text{id}_*(1)$, then we precisely have: $x \cdot 1_m = x$, $1_m \cdot x = x$, and $(x+y) \cdot (z+w) = x \cdot z + x \cdot w + y \cdot z + y \cdot w$, using unit and $\mathbb{Z}$ -bilinearity of the morphism. So, one-object category enriched over **Ab** with tensor product as monoidal product, is a unital ring.

Example 2.4. If consider the category of compactly generated spaces $\mathcal{U}$, then the product (i.e. the k -ification for original cartesian product) satisfies the following adjunction:

$$\text{Hom}_{\mathcal{U}}(X \times K, Y) \cong \text{Hom}_{\mathcal{U}}(X, Y^K)$$

where $Y^K := \text{Hom}_{\mathcal{U}}(K, Y)$ Which, it's actually cartesian closed and cocomplete.

Example 2.5. If $\mathcal{V}$ is monoidally closed + cocomplete, then its internal hom functor $\mathcal{V}(-, -)$ induces a structure where $\mathcal{V}$ is self-enriched.

The unit morphism is identified by the adjunction $\text{Hom}_{\mathcal{V}}(* \times v, v) \cong \text{Hom}_{\mathcal{V}}(*, \mathcal{V}(v, v))$, identified as the image of the natural isomorphism $\gamma_v : * \times v \xrightarrow{\sim} v$, denoted as $\gamma_v^\# : * \rightarrow \mathcal{V}(v, v)$.

Before building up the composition morphism, we need to consider an *evaluation morphism* $\epsilon : \mathcal{V}(u, v) \times u \rightarrow v$. This is constructed using the same adjunction, together with the pullback of identity $1 : \mathcal{V}(u, v) \xrightarrow{\sim} \mathcal{V}(u, v)$, or $\epsilon := 1^\flat$. Then, we define the pushforward of the following morphism:

$$\phi : \mathcal{V}(v, w) \times \mathcal{V}(u, v) \times u \xrightarrow{\text{id} \times \epsilon} \mathcal{V}(v, w) \times v \xrightarrow{\epsilon} w, \quad \circ := \phi^\# : \mathcal{V}(v, w) \times \mathcal{V}(u, v) \rightarrow \mathcal{V}(u, w)$$

3 2-Categories and Bicategories

Finally, to define a 2-category, there are multiple ways of defining it:

Definition 3.1. A category $\mathcal{C}$ is a *2-category*, if it's an ∞ -category, such that all 3-morphism is an equivalence, and all $n \geq 3$ -morphisms that're parallel are equivalent.

If anyone can think in an ∞ -category manner then they can take this definition, at least I can't...

The first (and most intuitive) definition of a 2-category uses enrichment:

Definition 3.2. A category $\mathcal{C}$ is a *2-category*, if the hom-functor provides an enrichment over $\mathbf{Cat}$, the category of small categories.

This means, each $X, Y \in \mathcal{C}$ has $\text{Hom}_{\mathcal{C}}(X, Y)$ actually forms a small category (called a *hom-category*), and the composition of morphisms $\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z)$ is a functor.

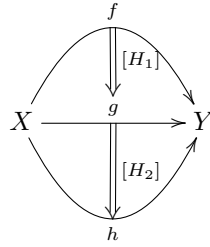
Here, we call the elements in $\text{Hom}_{\mathcal{C}}(X, Y)$ the *1-cells*, and the morphisms within the hom-category are called the *2-cells*.

Example 3.3. If consider a 2-category under this definition with only 1-object, then the corresponding hom-category $\mathcal{A} := \text{Hom}(*, *)$ has an associative “composition” $\circ : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$, and the existence of identity morphism serves as an actual unit under this composition.

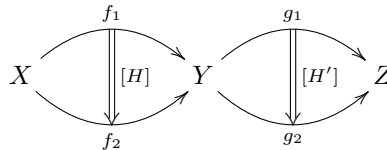
This realizes $\mathcal{A}$ as a strict monoidal category (meaning the monoidal product is strictly associative, similar for unit). If add an extra requirement that $\text{Hom}(*, *)$ is commutative, then it becomes a strict symmetric monoidal category.

For another instance, recalling back to our initial example of $\mathbf{Top}$, and each $\text{Hom}_{\mathbf{Top}}(X, Y)$ realized as a groupoid using homotopic classes of homotopies as morphisms between maps, this is a well-defined 2-category structure on $\mathbf{Top}$. Here, I want to emphasize an aspect about the 2-cells, since they have 2 different “compositions”:

Example 3.4. Fixing two spaces X, Y , for any three maps $f, g, h \in \text{Hom}_{\mathbf{Top}}(X, Y)$, pick two classes of homotopies $[H_1] : f \Rightarrow g$, $[H_2] : g \Rightarrow h$, then the concatenation $[H_2 \cdot H_1] : f \Rightarrow h$ is still a morphism within $\text{Hom}_{\mathbf{Top}}(X, Y)$. This is a composition while fixing a hom-category, called *vertical composition*:



Then, given $f_1, f_2 : X \rightarrow Y$, $g_1, g_2 : Y \rightarrow Z$, together with $[H] : f_1 \Rightarrow f_2$ and $[H'] : g_1 \Rightarrow g_2$ two homotopies, the composition functor states the existence of the 2-cell $[H'] \circ [H] : g_1 \circ f_1 \Rightarrow g_2 \circ f_2$. This is a composition over different hom-categories, called *horizontal composition*:



Classically, this can be written as composition of homotopies, restricting to the same time parameter:

$$H' * H : X \times I \rightarrow Z, \quad (H' * H)(x, t) := H'(H(x, t), t)$$

Similar vertical / horizontal compositions of natural transformations can also be realized in category of small categories.

Definition 3.5. Given $\mathcal{C}$ a 2-category, the *vertical composition* of 2-cells is the composition within each hom-category $\text{Hom}_{\mathcal{C}}(X, Y)$.

The *horizontal composition* of 2-cells is the one induced by the composition functor $\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z)$: Given $H : f_1 \Rightarrow f_2$ in $\text{Hom}_{\mathcal{C}}(X, Y)$ and $H' : g_1 \Rightarrow g_2$ in $\text{Hom}_{\mathcal{C}}(Y, Z)$, define their horizontal composition as $H' * H := \circ(H', H) : g_1 \circ f_1 \Rightarrow g_2 \circ f_2$.

Here, functoriality guarantees the two compositions of 2-cells commute, as horizontal is described by functors from product category (of hom-categories), vertical is described by compositions within each component of product category.

However, it turns out this definition is too restrictive, partially because when working on a functor level, requiring the functors to be strictly equal doesn't apply to most applications. So, an alternative definition is defined, where 1 and 2-cells are now allowed to commute up to certain notion of isomorphisms:

Definition 3.6. A *bicategory* is a tuple $(\mathcal{B}, 1, c, a, \ell, r)$, such that:

- The objects $\text{Ob}(\mathcal{B})$ is a class, called *0-cells* (also denoted $\mathcal{B}_0$).
- The *morphism/hom categories*: For each $X, Y \in \mathcal{B}$, there exists a category $\mathcal{B}(X, Y)$, with the objects called *1-cells* (denoted $\mathcal{B}_1$), and the morphisms are called *2-cells* (denoted $\mathcal{B}_2$).

I. Terminologies for 2-cells: The composition in each $\mathcal{B}(X, Y)$ is called *vertical compositions* (here, denote use $\odot$ to denote it), the identity morphisms in $\mathcal{B}(X, Y)$ are called *identity 2-cells*, while an isomorphism is called *invertible 2-cells*, with inverse called *vertical inverse*.

II. Horizontal Composition: For each 0-cells $X, Y, Z \in \mathcal{B}_0$, here exists a *horizontal composition functor*

$$c_{XYZ} : \mathcal{B}(Y, Z) \times \mathcal{B}(X, Y) \rightarrow \mathcal{B}(X, Z)$$

Where given 1-cells $f \in \mathcal{B}(X, Y)$, $g \in \mathcal{B}(Y, Z)$, and 2-cells $\alpha \in \mathcal{B}(X, Y)$, $\beta \in \mathcal{B}(Y, Z)$, we denote

$$c_{XYZ}(g, f) = g \circ f, \quad c_{XYZ}(\beta, \alpha) = \beta * \alpha$$

III. Associator: Given $W, X, Y, Z \in \mathcal{B}_0$, there exists *associator isomorphism*

$$a_{WXYZ} : c_{WXZ} \circ (c_{XYZ} \times \text{Id}_{\mathcal{B}(W, X)}) \xrightarrow{\sim} c_{WYZ} \circ (\text{Id}_{\mathcal{B}(Y, Z)} \times c_{WYX})$$

between the functors $\mathcal{B}(Y, Z) \times \mathcal{B}(X, Y) \times \mathcal{B}(W, X) \rightarrow \mathcal{B}(W, Z)$ (this is saying, composition of 1-cells are only up to isomorphism, and may be different from the hom class of $\mathcal{B}_0$).

IV. Identity 1-cells: Given any 0-cell $X \in \mathcal{B}_0$, there exists a functor $\eta_X : * \rightarrow \mathcal{B}(X, X)$, with image denoted as id_X . Which, there is the natural isomorphism between the following pairings:

$$\begin{array}{ccc}
* \times \mathcal{B}(X, Y) & \xrightarrow{\eta_Y \times \text{Id}_{\mathcal{B}(X, Y)}} & \mathcal{B}(Y, Y) \mathcal{B}(X, Y) \xrightarrow{c_{XY Y}} \mathcal{B}(X, Y) \\
& & \downarrow \ell_{XY} \\
* \times \mathcal{B}(X, Y) & \xrightarrow{\text{Id}_{\mathcal{B}(X, Y)}} & \mathcal{B}(X, Y) \\
& & \uparrow r_{XY} \\
\mathcal{B}(X, Y) \times * & \xrightarrow{\text{Id}_{\mathcal{B}(X, Y)} \times \eta_X} & \mathcal{B}(Y, X) \mathcal{B}(X, X) \xrightarrow{c_{X X Y}} \mathcal{B}(X, Y)
\end{array}$$

Here, ℓ_{XY} is called the *left unitor*, conversely r_{XY} is the *right unitor*.

Some axioms it satisfies include:

- *Unit:* Given $f \in \mathcal{B}(X, Y)$, $\text{id}_Y \in \mathcal{B}(Y, Y)$, and $g \in \mathcal{B}(Y, Z)$, the following diagram holds:

$$\begin{array}{ccc}
(g \circ \text{id}_Y) \circ f & \xrightarrow{a} & g \circ (\text{id}_Y \circ f) \\
& \searrow r_{YZ} * \text{id} & \swarrow \text{id} * \ell_{XY} \\
& g \circ f &
\end{array}$$

- *Pentagon Axiom:* Given 1-cells f, g, h, ℓ that are composable in order, the following diagram holds:

$$\begin{array}{ccccc}
& & (\ell \circ h) \circ (g \circ f) & & \\
& \nearrow a_{k \circ h, g, f} & & \nwarrow a_{k, h, g \circ f} & \\
((k \circ h) \circ g) \circ f & & & & k \circ (h \circ (g \circ f)) \\
\downarrow a_{k, j, g} * \text{id}_f & & & & \uparrow \text{id}_k * a_{h, g, f} \\
(k \circ (h \circ g)) \circ f & \xrightarrow{a_{k, h \circ g, f}} & k \circ ((h \circ g) \circ f) & &
\end{array}$$

As a canonical example:

Example 3.7. A one-object bicategory is a monoidal category; now the composition is associative up to natural isomorphism, the unit and pentagon axioms precisely matches the axioms for monoidal category.

More generally, given a bicategory $\mathcal{B}$, then any object $X \in \mathcal{B}$ has its hom-category $\mathcal{B}(X, X)$ being a monoidal category.

Example 3.8. Now, we consider the *category of bimodules*:

We let objects be the commutative rings (so the underlying category is $\mathbf{CRing}$), and for two rings R, S , the corresponding hom-category be the category of bimodules $\mathbf{BiMod}(R, S)$ (so 1-cells are (R, S) -bimodules), with 2-cells being bimodule homomorphisms.

Then, given (R, S) -bimodule M and (S, T) -bimodule N , we define the composition of 1-cells as *S-tensor* $\otimes_S : \mathbf{BiMod}(S, T) \times \mathbf{BiMod}(R, S) \rightarrow \mathbf{BiMod}(R, T)$ by $(N, M) \mapsto N \otimes_S M$, here the functoriality of composition follows from tensor product.

The associator between composition follows from the isomorphism of tensor products across different commutative rings: Given (R, S) -bimodule M , (S, T) -bimodule N , and (T, L) -bimodule P , there is an isomorphism as follow (by changing the paranthesis):

$$(P \otimes_T N) \otimes_S M \cong P \otimes_T (N \otimes_S M), \quad (p \otimes n) \otimes m \mapsto p \otimes (n \otimes m)$$

Such isomorphism is universal, hence form a natural isomorphism between multiple compositions.

4 Appendix: Category of Based Objects

Let $\mathcal{V}$ be a cartesian closed monoidal category that's also bicomplete (so, categorical product $\times$ is our monoidal product, final object $*$ $\in \mathcal{V}$ serves as unit due to the fact $* \times X \cong X$ by checking universality; it has an internal hom functor $\mathcal{V}(-, -) : \mathcal{V}^{\text{op}} \times \mathcal{V} \rightarrow \mathcal{V}$ generating the adjoint to $\times$ when fixing one variable, while all desired limits / colimits exist).

The goal is to generate something similar to the *category of based objects* analogous to the category of based sets / based spaces over **Set** / **Top** respectively, with *smash product* as our monoidal product. Let's first check these two categories for an understanding:

Example 4.1. Given **Set** or **Top**, the category of based spaces is equivalent to the slice category of the final object $*$ $\in \mathbf{Top}$ (the image of $*$ becomes the basepoint). We have each space $(X, x_0), (Y, y_0)$ satisfies wedge product $X \vee Y := X \sqcup Y x_0 \sim y_0$, and smash product $X \wedge Y := X \times Y X \vee Y$ using the identification with basepoint.

Consider the inclusion $X \rightarrow X \times Y$ by $x \mapsto (x, y_0)$, and $Y \rightarrow X \times Y$ by $y \mapsto (x_0, y)$, this generates a map $X \sqcup Y \rightarrow X \times Y$. Which, we claim the following fibre coproduct diagram is satisfied:

$$\begin{array}{ccc}
 X \sqcup Y & \longrightarrow & X \times Y \\
 \downarrow & & \downarrow \\
 * & \longrightarrow & X \wedge Y \\
 & \searrow & \downarrow \\
 & & Z
 \end{array}$$

By sending $*$ to the class of $X \vee Y$ in $X \wedge Y$, since the image of $X \sqcup Y$ in $X \times Y$ is precisely $X \vee Y$, the above diagram commutes. Moreover, if maps into Z makes the above diagram commutes, we must have the image of $X \vee Y \subseteq X \times Y$ be the point specified by $*$, hence must uniquely factor through $X \wedge Y$ using the quotient space / set property.

Also, if consider the spaces of based maps $F(X, Y)$, notice given $s_X : * \rightarrow X$ and $s_Y : * \rightarrow Y$ two structure maps, it satisfies the following fibre product property:

$$\begin{array}{ccc}
 Z & & \\
 \searrow & \swarrow & \\
 F(X, Y) & \xrightarrow{\iota} & \text{Hom}_{\mathbf{Top}}(X, Y) \\
 \downarrow & & \downarrow \circ s_X \\
 \text{Hom}_{\mathbf{Top}}(*, *) & \xrightarrow{s_Y \circ -} & \text{Hom}_{\mathbf{Top}}(*, Y)
 \end{array}$$

With the examples in mind, we can construct the generalization:

Construction 4.2. Consider the slice category of the final object $\mathcal{V}_*$, then for any $X, Y \in \mathcal{V}_*$, the product is naturally in the slice category also:

$$\begin{array}{ccc}
 & * & \\
 & \downarrow & \\
 X & \longleftarrow X \times Y \longrightarrow & Y
 \end{array}$$

To mimic the morphism $X \sqcup Y \rightarrow X \times Y$, we consider morphisms $X \rightarrow X \times Y$ (and respectively, for $Y \rightarrow X \times Y$) defined as follow (for Y , switch up the X and Y):

$$\begin{array}{ccccc}
 & & X & \longrightarrow & * \\
 & \swarrow 1_X & \downarrow \exists! & & \downarrow \\
 X & \longleftarrow & X \times Y & \longrightarrow & Y
 \end{array}$$

So, after generating these two morphisms, one naturally induces $X \sqcup Y \rightarrow X \times Y$. So, we can define the *smash product* $X \wedge Y$ as the fibre coproduct as follow:

$$\begin{array}{ccc}
 X \sqcup Y & \longrightarrow & X \times Y \\
 \downarrow & & \downarrow \\
 * & \longrightarrow & X \wedge Y \\
 \searrow & & \swarrow \\
 & & Z
 \end{array}$$

Here, need to emphasize that this construction is not necessarily symmetric monoidal (as there are counterexamples in $\mathbf{Top}$ that violates associativity). But, once this is associative (up to isomorphism), the universal property establishes $-\wedge -: \mathcal{V}_* \times \mathcal{V}_* \rightarrow \mathcal{V}_*$ as a functor. Moreover, the universal property can be used to establish an internal hom functor $\mathcal{V}_*(-, -) : \mathcal{V}_* \times \mathcal{V}_* \rightarrow \mathcal{V}_*$ as the fibre product diagram:

$$\begin{array}{ccc}
 Z & & \\
 \searrow & \swarrow & \\
 & \mathcal{V}_*(v, u) \longrightarrow \mathcal{V}(v, u) & \\
 & \downarrow & \downarrow \\
 & \mathcal{V}(*, *) \longrightarrow \mathcal{V}(*, u) &
 \end{array}$$

This is a generalization of category of based spaces to arbitrary cartesian closed symmetric monoidal category. As an application, apply to the category of small categories $\mathbf{Cat}$, it picks out an object as a *basepoint* for each category. Then, the internal hom functor is really the functors preserving the basepoint, which matches our understanding of based spaces.